

Edward Francisco Buccieri

ResearchGate: [Edward-Francisco-Buccieri](#) | LinkedIn: [edward-francisco-buccieri](#)

Summary & Research Interests

Graduate mathematician specializing in Optimal Transport and Theoretical Machine Learning, with applied experience in digital pathology via deep learning. Looking ahead to my PhD, I aim to further explore the applications of Wasserstein Gradient Flows within Machine Learning.

Education

M.Sc. in Mathematics

Università degli Studi di Pavia

Date: December 2023 – July 2026

Grade: 110/110 *cum Laude*

Thesis: *Gradient Flows and Optimal Transport in Neural Network Training*

Advisor: Prof. Matteo Negri

Key Topics: Wasserstein spaces, JKO Scheme, Mean-Field Limits, Particle Gradient Flows.

B.Sc. in Mathematics

Università degli Studi di Milano-Bicocca

Date: Graduated November 2023

Thesis: *Magnetic Monopoles on Projective Spaces*

Advisor: Prof. Alessandro Tomasiello

Research Experience

Researcher

Centro di Malattie Genetiche Cardiovascolari, Policlinico San Matteo

Location: Pavia, Italy

Date: June 2024 – Present

Description: Focus on the application of deep learning to digital pathology.

Conference Presentations & Published Abstracts:

- **ePoster:** *"A comparative deep learning approach using foundation models for identification of acute cellular rejection in endomyocardial biopsies of heart transplant recipients."* ESC Congress, August 2026, Munich.
- **Oral Presentation:** *"A Novel AI-Based Digital Pathology Pipeline for Acute Cellular Rejection Assessment in Endomyocardial Biopsies from heart transplanted patients."* Conoscere e Curare il Cuore (CCC), March 2026, Florence.
- **ePoster & Published Abstract:** *"AI-Based Identification of Acute Cellular Rejection in Digital Pathology Images of Endomyocardial Biopsies from heart transplanted patients."* ESC Congress, August 2025, Madrid. Abstract published in *European Heart Journal*, 46(Supplement_1), November 2025. DOI: [10.1093/eurheartj/ehaf784.4401](https://doi.org/10.1093/eurheartj/ehaf784.4401).

Technical Skills

– **Programming Languages:** Python, C, Java, MATLAB.

- **Machine Learning & Tools:** PyTorch, TensorFlow, GeomLoss.
- **Formatting & Typesetting:** LaTeX.

Summer Schools & Workshops

- **Mathematical Foundations of Deep Learning** *Aug 3–7, 2026*
University of Cambridge, UK. Rigorous education in the foundational mathematics of modern AI.
- **Mathematical Foundations of Machine Learning (MFML2026)** *May 25–29, 2026*
Hotel Montana, Monte Bondone (TN), Italy. Intensive school in mathematical foundations of ML.
- **Theoretical Foundations of Machine Learning (TFML2025)** *June 23–27, 2025*
MaLGa Center, University of Genoa, Italy. Summer school in theoretical Machine Learning.

Extracurricular Activities

- **Student Representative** *June 2024 – June 2026*
Università degli Studi di Pavia. Actively represented student interests in academic and administrative matters.
- **Volunteer Member** *Apr 2017 – May 2019*
ForumGiovani Gessate, Italy. Assisted children with homework and helped organize local events.

Language Skills

- **Italian:** Native Speaker
- **English:** CEFR C1 (TOEFL iBT 5.0/6)